

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP

☒ Static

192.50.20.10

255.255.255.0

192.50.20.1

0.0.0.0

IPv6 Configuration

☒ Automatic

☒ Static

FE80::260:2FFF:FEE1:80A9

802.1X

☒ Use 802.1X Security

Physical Config Desktop Programming Attributes

IP Configuration

X

Interface FastEthernet0

▼

IP Configuration

☒ DHCP☒ Static

172.36.0.10

255.255.0.0

172.36.0.1

0.0.0.0

IPv6 Configuration

☒ Automatic☒ Static

FE80::201:C9FF:FEA5:C5A3

802.1X

☒ Use 802.1X Security

Authentication MD5

✓

Username

Password

Physical Config Desktop Programming Attributes

IP Configuration

✕

Interface FastEthernet0

▼

IP Configuration

☒ DHCP ☒ Static

223.0.50.10

255.255.255.0

223.0.50.1

0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

FE80::250:FFF:FE59:85A7

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Physical Config CLI Attributes

```
Router1>
Router1>en
Router1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.50.20.1	YES	manual	up	up
FastEthernet0/1	12.0.0.1	YES	manual	up	up

```
Router1#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
C    12.0.0.0/8 is directly connected, FastEthernet0/1
S    23.0.0.0/8 [1/0] via 12.0.0.2
S    24.0.0.0/8 [1/0] via 12.0.0.2
S    26.0.0.0/8 [1/0] via 12.0.0.2
S    32.0.0.0/8 [1/0] via 12.0.0.2
S    45.0.0.0/8 [1/0] via 12.0.0.2
S    172.0.0.0/16 [1/0] via 12.0.0.2
S    172.36.0.0/16 [1/0] via 12.0.0.2
C    192.50.20.0/24 is directly connected, FastEthernet0/0
S    223.0.0.0/24 [1/0] via 12.0.0.2
S    223.0.50.0/24 [1/0] via 12.0.0.2
```

```
Router1#
```

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%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up

Router2>

Router2>

Router2>

Router2>

Router2>en

Router2#

Router2#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	12.0.0.2	YES	manual	up	up
FastEthernet0/1	23.0.0.1	YES	manual	up	up
FastEthernet1/0	26.0.0.1	YES	manual	up	up
FastEthernet1/1	24.0.0.1	YES	manual	up	up

Router2#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
C    12.0.0.0/8 is directly connected, FastEthernet0/0
C    23.0.0.0/8 is directly connected, FastEthernet0/1
C    24.0.0.0/8 is directly connected, FastEthernet1/1
C    26.0.0.0/8 is directly connected, FastEthernet1/0
S    32.0.0.0/8 [1/0] via 23.0.0.2
      [1/0] via 24.0.0.2
S    45.0.0.0/8 [1/0] via 23.0.0.2
      [1/0] via 24.0.0.2
S    172.36.0.0/16 [1/0] via 23.0.0.2
      [1/0] via 24.0.0.2
S    192.50.20.0/24 [1/0] via 12.0.0.1
S    223.0.50.0/24 [1/0] via 26.0.0.2
```

--More--

Physical Config CLI Attributes

```
Router3>
Router3>
Router3>
Router3>
Router3>en
Router3#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          23.0.0.2        YES manual up          up
FastEthernet0/1          32.0.0.1        YES manual up          up
Router3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
C    23.0.0.0/8 is directly connected, FastEthernet0/0
S    24.0.0.0/8 [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
S    26.0.0.0/8 [1/0] via 32.0.0.2
      [1/0] via 23.0.0.1
C    32.0.0.0/8 is directly connected, FastEthernet0/1
S    45.0.0.0/8 [1/0] via 32.0.0.2
      [1/0] via 23.0.0.1
S    172.36.0.0/16 [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
S    192.50.20.0/24 [1/0] via 23.0.0.1
--More--
```

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```

Router4>
Router4>
Router4>
Router4>
Router4>en
Router4#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          24.0.0.2        YES manual up          up
FastEthernet0/1          45.0.0.1        YES manual up          up
Router4#
Router4#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 45.0.0.2
           [1/0] via 24.0.0.1
S    23.0.0.0/8 [1/0] via 24.0.0.1
           [1/0] via 45.0.0.2
C    24.0.0.0/8 is directly connected, FastEthernet0/0
S    26.0.0.0/8 [1/0] via 45.0.0.2
           [1/0] via 24.0.0.1
S    32.0.0.0/8 [1/0] via 24.0.0.1
           [1/0] via 45.0.0.2
C    45.0.0.0/8 is directly connected, FastEthernet0/1
S    172.36.0.0/16 [1/0] via 24.0.0.1
                  [1/0] via 45.0.0.2

--More--

```

```
Router5>
Router5>
Router5>en
Router5#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          172.36.0.1      YES manual up          up
FastEthernet0/1          32.0.0.2        YES manual up          up
FastEthernet1/0          45.0.0.2        YES manual up          up
FastEthernet1/1          unassigned      YES NVRAM   administratively down down
Router5#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 32.0.0.1
           [1/0] via 45.0.0.1
S    23.0.0.0/8 [1/0] via 32.0.0.1
           [1/0] via 45.0.0.1
S    24.0.0.0/8 [1/0] via 32.0.0.1
           [1/0] via 45.0.0.1
S    26.0.0.0/8 [1/0] via 32.0.0.1
           [1/0] via 45.0.0.1
C    32.0.0.0/8 is directly connected, FastEthernet0/1
C    45.0.0.0/8 is directly connected, FastEthernet1/0
C    172.36.0.0/16 is directly connected, FastEthernet0/0
S    192.50.20.0/24 [1/0] via 32.0.0.1
           [1/0] via 45.0.0.1
S    223.0.50.0/24 [1/0] via 32.0.0.1
           [1/0] via 45.0.0.1

Router5#
```


Physical Config CLI Attributes

```
Router6>
Router6>en
Router6#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          223.0.50.1      YES manual up          up
FastEthernet0/1          26.0.0.2        YES manual up          up
Router6#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 26.0.0.1
      [1/0] via 23.0.0.1
      [1/0] via 32.0.0.2
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 23.0.0.1
      [1/0] via 26.0.0.1
      [1/0] via 32.0.0.2
C    26.0.0.0/8 is directly connected, FastEthernet0/1
S    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
C    223.0.50.0/24 is directly connected, FastEthernet0/0

Router6#
```

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Command Prompt



Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time<1ms TTL=128

Reply from 223.0.50.10: bytes=32 time=1ms TTL=128

Reply from 223.0.50.10: bytes=32 time=1ms TTL=128

Reply from 223.0.50.10: bytes=32 time<1ms TTL=128

Ping statistics for 223.0.50.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=12ms TTL=120

Reply from 172.36.0.10: bytes=32 time=11ms TTL=124

Reply from 172.36.0.10: bytes=32 time=10ms TTL=120

Reply from 172.36.0.10: bytes=32 time=1ms TTL=124

Ping statistics for 172.36.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 12ms, Average = 8ms

C:\>ping 192.50.20.10

Pinging 192.50.20.10 with 32 bytes of data:

Reply from 192.50.20.10: bytes=32 time<1ms TTL=128

Reply from 192.50.20.10: bytes=32 time=8ms TTL=128

Reply from 192.50.20.10: bytes=32 time=8ms TTL=128

Reply from 192.50.20.10: bytes=32 time=8ms TTL=128

Ping statistics for 192.50.20.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 8ms, Average = 6ms

C:\>|

Command Prompt



```
Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=10ms TTL=120
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=10ms TTL=120
Reply from 223.0.50.10: bytes=32 time=10ms TTL=124

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 7ms

C:\>ping 192.50.20.10

Pinging 192.50.20.10 with 32 bytes of data:

Reply from 192.50.20.10: bytes=32 time=12ms TTL=120
Reply from 192.50.20.10: bytes=32 time<1ms TTL=124
Reply from 192.50.20.10: bytes=32 time=10ms TTL=120
Reply from 192.50.20.10: bytes=32 time=16ms TTL=124

Ping statistics for 192.50.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 9ms

C:\>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=33ms TTL=128
Reply from 172.36.0.10: bytes=32 time=1ms TTL=128
Reply from 172.36.0.10: bytes=32 time=8ms TTL=128
Reply from 172.36.0.10: bytes=32 time=8ms TTL=128

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 33ms, Average = 12ms

C:\>
```


Physical Config Desktop Programming Attributes

Command Prompt

Pinging 192.50.20.10 with 32 bytes of data:

Reply from 192.50.20.10: bytes=32 time=1ms TTL=125
Reply from 192.50.20.10: bytes=32 time=1ms TTL=125
Reply from 192.50.20.10: bytes=32 time=10ms TTL=125
Reply from 192.50.20.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.50.20.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=11ms TTL=120
Reply from 172.36.0.10: bytes=32 time=19ms TTL=124
Reply from 172.36.0.10: bytes=32 time=21ms TTL=120
Reply from 172.36.0.10: bytes=32 time=11ms TTL=124

Ping statistics for 172.36.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 11ms, Maximum = 21ms, Average = 15ms

C:\>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=11ms TTL=123
Reply from 223.0.50.10: bytes=32 time=8ms TTL=123
Reply from 223.0.50.10: bytes=32 time=8ms TTL=123
Reply from 223.0.50.10: bytes=32 time=8ms TTL=123

Ping statistics for 223.0.50.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 8ms, Maximum = 11ms, Average = 8ms

C:\>